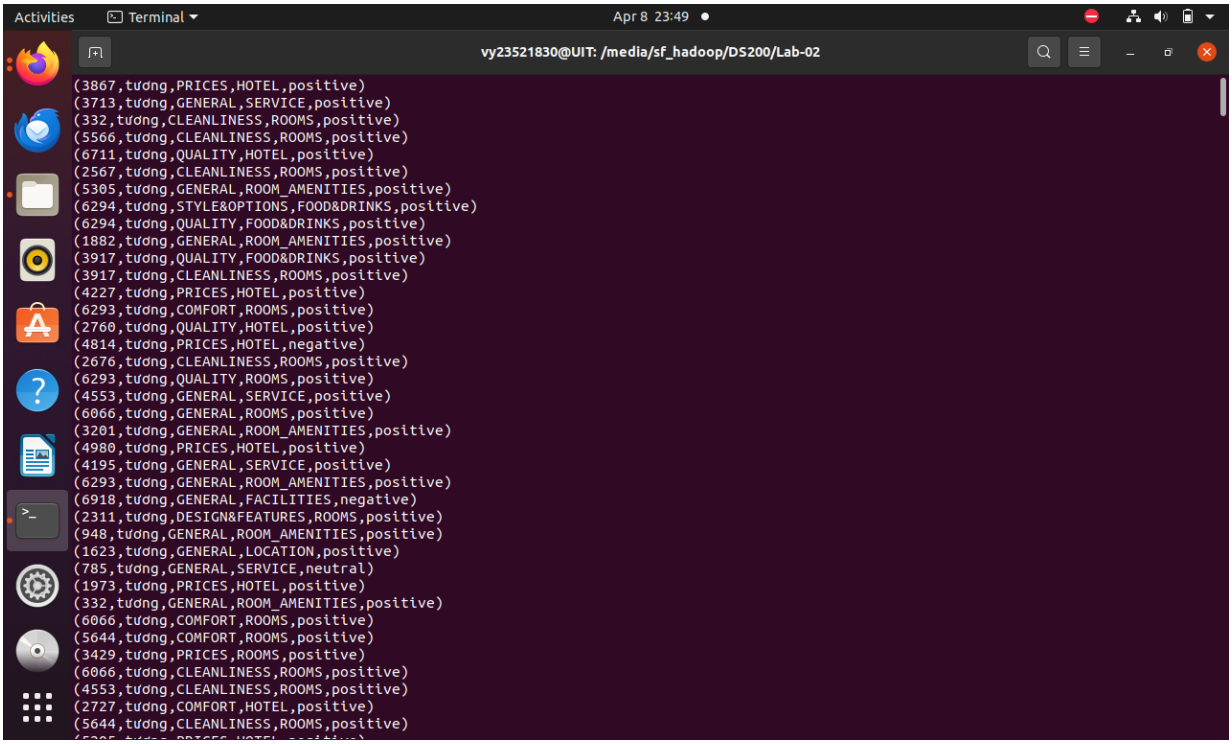
MSSV: 23521830

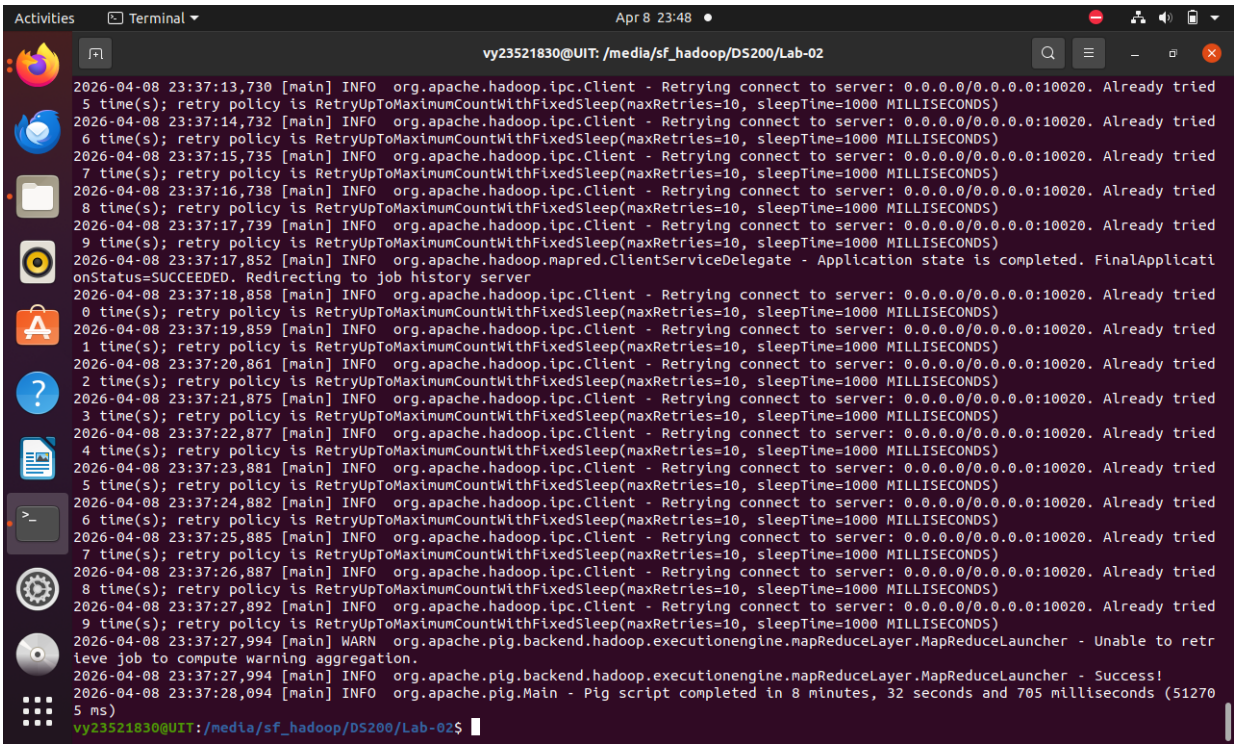
Họ và tên: Nguyễn Hoàng Vy

Lớp: DS200.Q21

Bài tập Thực hành 2

Câu 1:





Câu 2:

```
2026-04-09 00:11:03,997 [main] INFO  org.apache.pig.Main - Pig script completed in 4 minutes, 9 seconds and 241 milliseconds (249241
 ms)
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-02$ pig -x mapreduce bai2.pig
2026-04-09 00:11:59,059 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-04-09 00:11:59,063 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-04-09 00:11:59,063 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-04-09 00:11:59,381 [main] INFO  org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-04-09 00:11:59,415 [main] INFO  org.apache.pig.Main - Logging error messages to: /media/sf_hadoop/DS200/Lab-02/pig_177566831937
1.log
2026-04-09 00:12:01,107 [main] INFO  org.apache.pig.impl.util.Utils - Default bootup file /home/vy23521830/.pigbootup not found
2026-04-09 00:12:01,827 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, u
se mapreduce.jobtracker.address
2026-04-09 00:12:01,828 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file syst
em at: hdfs://localhost:9000
2026-04-09 00:12:04,176 [main] INFO  org.apache.pig.PigServer - Pig Script ID for the session: PIG-bai2.pig-8030324e-c62e-4b0e-96c5-
cc529238fbbd
2026-04-09 00:12:04,181 [main] WARN  org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-04-09 00:12:07,656 [main] WARN  org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_LONG 1 time(s).
2026-04-09 00:12:07,709 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-04-09 00:12:07,784 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate
 code.
2026-04-09 00:12:07,986 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, Col
umnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, Nes
tedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTy
peCastInserter]}
2026-04-09 00:12:08,291 [main] INFO  org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (Tenured Gen) of size 699072512
 to monitor. collectionUsageThreshold = 489350752, usageThreshold = 489350752
2026-04-09 00:12:08,759 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation th
reshold: 100 optimistic? false
2026-04-09 00:12:08,829 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move alg
ebraic foreach to combiner
2026-04-09 00:12:08,919 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size
 before optimization: 1
2026-04-09 00:12:08,919 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size
 after optimization: 1
2026-04-09 00:12:09,133 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
```

```
job_1775664781237_0017  1       1       15      15      15      15      15      15      15      15      aspect_count,aspect_group,ra
w_data  GROUP_BY,COMBINER       /Lab02_DS/output_bai2_3,

Input(s):
Successfully read 11770 records (1861195 bytes) from: "/Lab02_DS/hotel-review.csv"

Output(s):
Successfully stored 7 records (99 bytes) in: "/Lab02_DS/output_bai2_3"

Counters:
Total records written : 7
Total bytes written : 99
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1775664781237_0017


2026-04-09 00:23:03,652 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 00:23:03,685 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 00:23:03,829 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 00:23:03,880 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 00:23:03,991 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 00:23:03,999 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 00:23:04,085 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-09 00:23:04,179 [main] INFO  org.apache.pig.Main - Pig script completed in 11 minutes, 5 seconds and 199 milliseconds (66519
9 ms)
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-02$
```

Câu 3:

```
Activities  Terminal  Apr 9 00:44
vy23521830@UIT: /media/sf_hadoop/DS200/Lab-02
vy23521830@UIT: /media/sf_hadoop/DS200/Lab-02   vy23521830@UIT: /media/sf_hadoop/DS200/Lab-02
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-02$  pig -x mapreduce bai3.pig
2026-04-09 00:43:17,052 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-04-09 00:43:17,056 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-04-09 00:43:17,056 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-04-09 00:43:17,279 [main] INFO  org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-04-09 00:43:17,280 [main] INFO  org.apache.pig.Main - Logging error messages to: /media/sf_hadoop/DS200/Lab-02/pig_177567019724
0.log
2026-04-09 00:43:18,132 [main] INFO  org.apache.pig.impl.util.Utils - Default bootup file /home/vy23521830/.pigbootup not found
2026-04-09 00:43:18,400 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, u
se mapreduce.jobtracker.address
2026-04-09 00:43:18,400 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file syst
em at: hdfs://localhost:9000
2026-04-09 00:43:19,642 [main] INFO  org.apache.pig.PigServer - Pig Script ID for the session: PIG-bai3.pig-d46f24bf-14f3-4dd0-b84f-
b425207ec81c
2026-04-09 00:43:19,642 [main] WARN  org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-04-09 00:43:21,229 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,
FILTER,LIMIT
2026-04-09 00:43:21,451 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate
 code.
2026-04-09 00:43:21,628 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, Col
umnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, Nes
tedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTy
peCastInserter]}
2026-04-09 00:43:21,882 [main] INFO  org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (Tenured Gen) of size 699072512
 to monitor. collectionUsageThreshold = 489350752, usageThreshold = 489350752
2026-04-09 00:43:22,043 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation th
reshold: 100 optimistic? false
2026-04-09 00:43:22,219 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move alg
ebraic foreach to combiner
2026-04-09 00:43:22,289 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Se
condary Key Optimization for MapReduce node scope-37
2026-04-09 00:43:22,336 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size
 before optimization: 4
2026-04-09 00:43:22,336 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size
 after optimization: 4
2026-04-09 00:43:22,701 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
```

```
Activities  Terminal  Apr 9 01:03
vy23521830@UIT: /media/sf_hadoop/DS200/Lab-02
2026-04-09 01:03:53,465 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:03:53,522 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:03:53,547 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:03:53,605 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:03:53,625 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:03:53,713 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:03:53,721 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:03:53,824 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:03:53,847 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:03:53,913 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:03:53,948 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:03:54,028 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:03:54,045 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:03:54,103 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:03:54,123 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:03:54,163 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:03:54,175 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:03:54,203 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-09 01:03:54,271 [main] INFO  org.apache.pig.Main - Pig script completed in 4 minutes, 20 seconds and 437 milliseconds (26043
7 ms)
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-02$
```

Câu 4:

```
Activities  Terminal ▾                    Apr 9 01:07 ●
vy23521830@UIT: /media/sf_hadoop/DS200/Lab-02
7 ms)
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-02$ pig -x mapreduce bai4.pig
2026-04-09 01:06:04,971 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-04-09 01:06:04,986 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-04-09 01:06:05,003 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-04-09 01:06:05,223 [main] INFO  org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-04-09 01:06:05,252 [main] INFO  org.apache.pig.Main - Logging error messages to: /media/sf_hadoop/DS200/Lab-02/pig_177567156521
5.log
2026-04-09 01:06:06,140 [main] INFO  org.apache.pig.impl.util.Utils - Default bootup file /home/vy23521830/.pigbootup not found
2026-04-09 01:06:06,387 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, u
se mapreduce.jobtracker.address
2026-04-09 01:06:06,391 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file syst
em at: hdfs://localhost:9000
2026-04-09 01:06:07,249 [main] INFO  org.apache.pig.PigServer - Pig Script ID for the session: PIG-bai4.pig-38061509-dbaf-4ddd-9477-
f43a2a35ae00
2026-04-09 01:06:07,250 [main] WARN  org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-04-09 01:06:08,615 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is depreca
ted. Instead, use mapreduce.output.textoutputformat.separator
2026-04-09 01:06:08,647 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-04-09 01:06:08,705 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate
 code.
2026-04-09 01:06:08,753 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, Col
umnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, Nes
tedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTy
peCastInserter]}
2026-04-09 01:06:09,047 [main] INFO  org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (Tenured Gen) of size 699072512
 to monitor. collectionUsageThreshold = 489350752, usageThreshold = 489350752
2026-04-09 01:06:09,242 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation th
reshold: 100 optimistic? false
2026-04-09 01:06:09,303 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move alg
ebraic foreach to combiner
2026-04-09 01:06:09,349 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move alg
ebraic foreach to combiner
2026-04-09 01:06:09,383 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Se
condary Key Optimization for MapReduce node scope-82
2026-04-09 01:06:09,385 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Se
condary Key Optimization for MapReduce node scope-87
2026-04-09 01:06:09,417 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size
```

```
Activities  Terminal ▾                    Apr 9 01:08 ●
vy23521830@UIT: /media/sf_hadoop/DS200/Lab-02
2026-04-09 01:08:19,497 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:08:19,527 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:08:19,531 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:08:19,553 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:08:19,567 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:08:19,596 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:08:19,603 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:08:19,625 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:08:19,628 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:08:19,649 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:08:19,653 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:08:19,682 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:08:19,686 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:08:19,716 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:08:19,719 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:08:19,746 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:08:19,760 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:08:19,791 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-09 01:08:19,814 [main] INFO  org.apache.pig.Main - Pig script completed in 2 minutes, 14 seconds and 958 milliseconds (13495
8 ms)
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-02$
```

Câu 5:

```
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-02$ pig -x mapreduce bai5.pig
2026-04-09 01:12:28,901 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-04-09 01:12:28,909 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-04-09 01:12:28,909 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-04-09 01:12:29,068 [main] INFO  org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-04-09 01:12:29,075 [main] INFO  org.apache.pig.Main - Logging error messages to: /media/sf_hadoop/DS200/Lab-02/pig_177567194904
4.log
2026-04-09 01:12:29,648 [main] INFO  org.apache.pig.impl.util.Utils - Default bootup file /home/vy23521830/.pigbootup not found
2026-04-09 01:12:29,787 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, u
se mapreduce.jobtracker.address
2026-04-09 01:12:29,787 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file syst
em at: hdfs://localhost:9000
2026-04-09 01:12:30,516 [main] INFO  org.apache.pig.PigServer - Pig Script ID for the session: PIG-bai5.pig-3cf9a76d-e7b0-4bb4-a4cd-
7ea13384ae45
2026-04-09 01:12:30,517 [main] WARN  org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-04-09 01:12:31,444 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is depreca
ted. Instead, use mapreduce.output.textoutputformat.separator
2026-04-09 01:12:31,484 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY
2026-04-09 01:12:31,652 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate
 code.
2026-04-09 01:12:31,731 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, Col
umnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, Nes
tedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTy
peCastInserter]}
2026-04-09 01:12:31,965 [main] INFO  org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (Tenured Gen) of size 699072512
 to monitor. collectionUsageThreshold = 489350752, usageThreshold = 489350752
2026-04-09 01:12:32,063 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation th
reshold: 100 optimistic? false
2026-04-09 01:12:32,162 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move alg
ebraic foreach to combiner
2026-04-09 01:12:32,183 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Se
condary Key Optimization for MapReduce node scope-43
2026-04-09 01:12:32,199 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size
 before optimization: 2
2026-04-09 01:12:32,207 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size
 after optimization: 2
2026-04-09 01:12:32,402 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
```

```
Total bytes written : 243549
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1775664781237_0040  ->      job_1775664781237_0041,
job_1775664781237_0041


2026-04-09 01:13:38,263 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:13:38,308 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:13:38,363 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:13:38,395 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:13:38,495 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:13:38,530 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:13:38,660 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:13:38,686 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:13:38,747 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:13:38,751 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:13:38,812 [main] INFO  org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManage
r at /0.0.0.0:8032
2026-04-09 01:13:38,818 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-04-09 01:13:38,881 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-09 01:13:38,940 [main] INFO  org.apache.pig.Main - Pig script completed in 1 minute, 10 seconds and 136 milliseconds (70136
ms)
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-02$
```